Chapter 8: t-tests in R

Comparing Group Means with Confidence

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1 Overview

This reading prepares you for the lab session where you'll conduct **t-tests** in R. The t-test is one of the most widely used statistical tests in psychology, allowing us to compare means and determine whether observed differences are likely to reflect real effects or just random variation.

> **i What You'll Learn**
>
> By the end of this lab, you'll be able to:
>
> - Use `t.test()` to compare two groups (independent samples)
> - Use `t.test()` to compare paired measurements (paired samples)
> - Interpret t-test output including t-values, p-values, and confidence intervals
> - Choose the appropriate type of t-test for your research question

2 When to Use a t-test

The t-test answers the question: **Is the difference between two means larger than we'd expect by chance?**

Use a t-test when you have:

1. A **continuous** outcome variable (e.g., reaction time, mood score)
2. **Two groups** to compare OR two measurements from the same people
3. Reasonably **normally distributed** data (or a large enough sample)

2.1 Types of t-tests

Type	When to Use	Example
Independent samples	Different people in each group	Room A vs Room B
Paired samples	Same people measured twice	Pre vs Post mood
One sample	Compare to a known value	Sample mean vs population mean

3 The `t.test()` Function

R's built-in `t.test()` function handles all three types. The basic syntax is:

```
t.test(outcome ~ group, data = mydata)
```

Or for paired data:

```
t.test(x, y, paired = TRUE)
```

4 Independent Samples t-test

Use this when comparing **different people** in two groups.

4.1 Example: Comparing Rooms

Did Room A and Room B differ in reaction time?

```
# Compare reaction times between rooms
t.test(reaction_time ~ room, data = datalab)
```

4.2 Understanding the Output

```
    Welch Two Sample t-test

data:  reaction_time by room
t = -2.34, df = 45.2, p-value = 0.024
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -42.5  -3.2
sample estimates:
mean in group A   mean in group B
          265.3             288.1
```

Let's decode each part:

Output	Meaning
`t = -2.34`	The t-statistic; how many standard errors the difference is from 0
`df = 45.2`	Degrees of freedom (Welch's adjustment for unequal variances)
`p-value = 0.024`	Probability of this difference (or larger) if null hypothesis is true
`95 percent confidence interval`	Range of plausible values for the true difference

Output	Meaning
`sample estimates`	The actual group means

> **! Interpreting the p-value**
>
> A p-value of 0.024 means: **if there were truly no difference between rooms**, we'd see a difference this large (or larger) about 2.4% of the time. Since this is less than 0.05, we'd typically conclude the difference is statistically significant.

5 Paired Samples t-test

Use this when comparing **the same people** measured at two time points.

5.1 Example: Pre-Post Mood

Did mood change after DataLab?

```r
# Compare mood before and after
t.test(datalab$mood_pre, datalab$mood_post, paired = TRUE)
```

Or using formula notation with long-format data:

```r
t.test(mood ~ time, data = datalab_long, paired = TRUE)
```

5.2 Why Paired Tests Are More Powerful

Paired tests account for individual differences. Consider:

- Person A: Pre = 5, Post = 7 (change = +2)
- Person B: Pre = 8, Post = 10 (change = +2)

Both people increased by 2 points, even though their baseline scores differ. The paired test focuses on the **change within each person**, not the raw scores.

> **Rule of Thumb**
>
> If the same individuals provide both measurements, use a paired test. It's more sensitive because it removes between-person variability.

6 One Sample t-test

Use this to compare your sample mean to a specific value.

6.1 Example: Comparing to Chance

Is staring detection accuracy different from 50% (chance)?

```r
# Compare mean accuracy to 0.5 (50%)
t.test(datalab$staring_accuracy, mu = 0.5)
```

The `mu` argument specifies the value you're testing against.

7 Key Arguments for `t.test()`

Argument	Purpose	Default
`paired = TRUE/FALSE`	Paired or independent test	`FALSE`
`var.equal = TRUE/FALSE`	Assume equal variances?	`FALSE` (Welch's test)
`alternative`	Direction: `"two.sided"`, `"less"`, `"greater"`	`"two.sided"`
`mu`	Value to test against (one-sample)	`0`
`conf.level`	Confidence level	`0.95`

7.1 About Welch's t-test

By default, R uses **Welch's t-test**, which doesn't assume equal variances in the two groups. This is generally recommended as it's more robust.

If you have good reason to assume equal variances:

```r
t.test(reaction_time ~ room, data = datalab, var.equal = TRUE)
```

8 Extracting Specific Results

Sometimes you want to pull out specific values for reporting:

```r
# Run the test and save results
result <- t.test(reaction_time ~ room, data = datalab)

# Extract specific values
result$statistic    # t-value
result$parameter    # degrees of freedom
result$p.value      # p-value
result$conf.int     # confidence interval
result$estimate     # group means
```

8.1 Reporting Format (APA Style)

For your report, you might write:

Participants in Room A (M = 265.3 ms) had significantly faster reaction times than those in Room B (M = 288.1 ms), $t(45.2) = -2.34$, $p = .024$, 95% CI [−42.5, −3.2].

9 Effect Size: Cohen's d

The p-value tells you whether a difference is **statistically significant**, but not whether it's **practically meaningful**. For that, we use effect sizes.

Cohen's d expresses the difference in standard deviation units:

d value	Interpretation
0.2	Small effect
0.5	Medium effect
0.8	Large effect

You can calculate Cohen's d using the `effectsize` package:

```
library(effectsize)

# For independent samples
cohens_d(reaction_time ~ room, data = datalab)

# For paired samples
cohens_d(datalab$mood_pre, datalab$mood_post, paired = TRUE)
```

10 Common Errors and How to Fix Them

10.1 Error: grouping factor must have exactly 2 levels

```
# Wrong: variable has more than 2 groups
t.test(score ~ condition, data = mydata)
# Error if condition has levels: A, B, C
```

Fix: Filter to only two groups:

```
# Right: compare only A and B
filtered <- mydata |> filter(condition %in% c("A", "B"))
t.test(score ~ condition, data = filtered)
```

10.2 Error: not enough 'x' observations

Fix: Your data might have too many missing values. Check with:

```
sum(is.na(datalab$reaction_time))
```

10.3 Error: object 'datalab' not found

Fix: Make sure you've loaded your data first. In WebR, run the data loading code before analysis.

10.4 Paired test gives strange results

Fix: Make sure:

1. The two vectors have the same length
2. The observations are in the same order (person 1's pre matches person 1's post)
3. You included `paired = TRUE`

```r
# Check lengths match
length(datalab$mood_pre)
length(datalab$mood_post)
```

10.5 Warning: data are essentially constant

Fix: One or both groups have no variability (all values are the same). A t-test isn't appropriate here.

11 Assumptions of the t-test

11.1 1. Independence

Observations should be independent (one person's score doesn't influence another's). This is about study design, not something you can test in R.

11.2 2. Normality

The data should be approximately normally distributed, especially for small samples. Check with:

```r
# Visual check
hist(datalab$reaction_time)

# Or by group
library(ggplot2)
ggplot(datalab, aes(x = reaction_time)) +
  geom_histogram(bins = 15) +
  facet_wrap(~room)
```

> Robustness
>
> The t-test is quite robust to violations of normality, especially with larger samples (n > 30 per group). Don't worry too much unless your data are severely skewed.

11.3 3. Homogeneity of Variance (for independent samples)

Groups should have similar variability. Welch's t-test (the default) handles unequal variances, so this is less of a concern.

12 Preparation Checklist

Before arriving at the lab:

> ⚠ Before You Arrive
>
> - [ ] Read this chapter carefully
> - [ ] Review the lecture material on hypothesis testing
> - [ ] Understand the difference between paired and independent samples
> - [ ] Be able to interpret a p-value correctly
> - [ ] Have a device with a web browser ready (for WebR)

12.1 Key Questions to Consider

- What research questions from DataLab can we answer with t-tests?
- Which comparisons require paired tests vs independent tests?
- What would a significant result mean in practical terms?

12.2 What to Bring

- Laptop, tablet, or phone with a modern browser
- Notes on which DataLab variables you want to compare
- Questions about interpreting output

13 Summary

Concept	Key Point
t-test purpose	Compare means to see if difference is beyond chance
Independent samples	Different people in each group
Paired samples	Same people measured twice
p-value	Probability of data if null hypothesis is true
Effect size (d)	Practical magnitude of the difference

14 Quick Reference

```r
# Independent samples (different people)
t.test(outcome ~ group, data = mydata)
```

```r
# Paired samples (same people, two times)
t.test(pre, post, paired = TRUE)

# One sample (compare to known value)
t.test(scores, mu = 50)

# Extract results
result <- t.test(...)
result$p.value
result$conf.int
```

15 Additional Resources

- R Documentation for t.test — Official documentation
- ModernDive: Hypothesis Testing — Conceptual explanations
- Datanovia: t-test in R — Step-by-step tutorials with examples